

Week 4 — Session 3

Topic: Writing Efficient Code

Language: C++17

Focus: clean input handling, linear-time processing, and in-place array manipulation.

Task 1: Time Conversion

Practice: <https://www.hackerrank.com/challenges/time-conversion/problem>

Convert a 12-hour AM/PM time string into 24-hour format.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    string s;
    cin >> s;

    int hour = stoi(s.substr(0, 2));
    string ampm = s.substr(8, 2);

    if (ampm == "AM") {
        if (hour == 12) hour = 0;
    } else {
        if (hour != 12) hour += 12;
    }

    cout << setw(2) << setfill('0') << hour
         << s.substr(2, 6) << '\n';

    return 0;
}
```

Task 2: Move Zeroes

Practice: <https://leetcode.com/problems/move-zeroes/>

Move all zeroes to the end while maintaining the relative order of non-zero elements.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

void moveZeroes(vector<int>& nums) {
    int pos = 0;

    for (int x : nums) {
        if (x != 0)
            nums[pos++] = x;
    }

    while (pos < (int)nums.size())
        nums[pos++] = 0;
}

int main() {
    int n;
    cin >> n;

    vector<int> nums(n);
    for (int &x : nums) cin >> x;

    moveZeroes(nums);

    for (int x : nums)
        cout << x << ' ';
    cout << '\n';

    return 0;
}
```

All programs are standalone C++17 solutions and can be compiled with a standard C++17 compiler.